AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

SALES AND BUSINESS DATA CLEANING AND VISUALIZATION

Project Domain

Excel / Power BI

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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Year - 2026

3. Problem Statement

Organizations across industries collect large volumes of data on a daily basis. However, this data is often raw, inconsistent, incomplete, or poorly structured, making it difficult to derive meaningful insights. Decision-makers rely on accurate, clean, and well-visualized data to make informed choices, but without the right tools and skills, this data remains underutilized.

Additionally, the rapid growth of Artificial Intelligence and Generative AI tools such as Microsoft Copilot and AI Chatbots has created both an opportunity and a skills gap. Professionals who can effectively use these tools through well-structured prompts can significantly accelerate their work. However, most users lack the knowledge of prompt engineering techniques needed to unlock the full potential of these AI tools.

This project addresses these challenges by applying data cleaning, analysis, visualization, and AI integration techniques using industry-standard tools. The project will demonstrate how structured data workflows combined with AI assistance can produce accurate, actionable insights and outputs relevant to a chosen real-world domain.

Students should include in their write-up:

> The healthcare plays a crucial role in society by availability of medicines and medical products. Medical stores generate daily sales related to medicines, quantities sold prices and manufacturing information. This helps to plan for future sales and performance.
>
> Accurate sales data analysis is critical in the medical domain because it affect customer satisfaction and business profitability. Poor quality data can lead to stock shortages, financial losses. By analysis medical stores can improve there operational efficiency.
>
> Medical store owners, pharmacy managers, inventory managers, small healthcare businesses.

4. Objectives

List the specific goals your project aims to accomplish. Objectives should be clear and measurable.

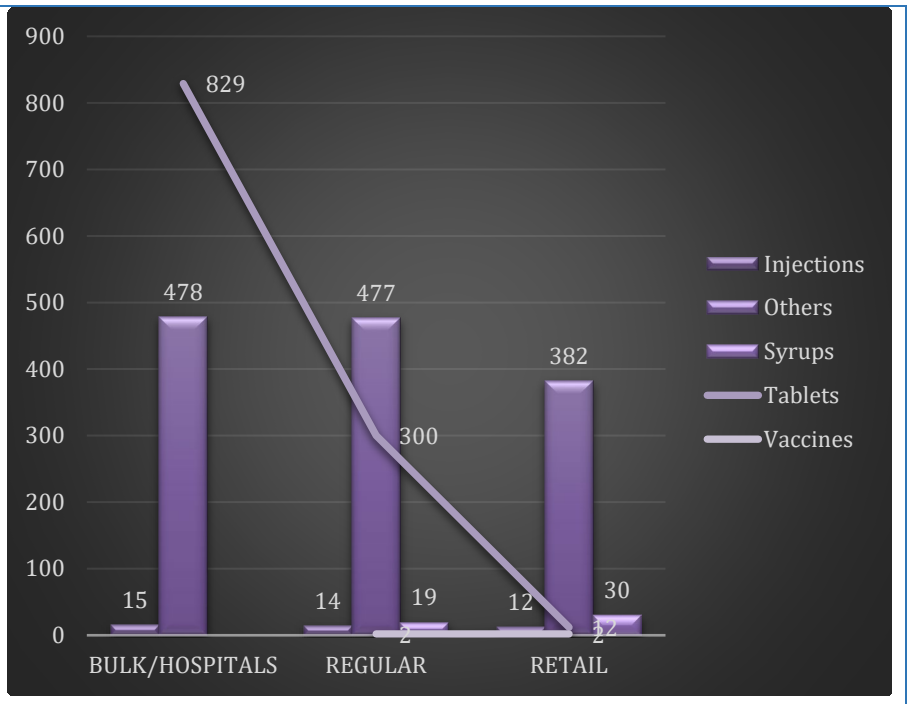
- To collect and explore the raw dataset, understand its structure, fields, and data quality issues.
- To clean and preprocess the data by removing duplicates, handling missing values, correcting formats, and standardizing fields.
- To apply domain-relevant analysis using the assigned tool (Excel formulas and pivot tables, Power BI measures, or AI prompt techniques).

- To build an interactive dashboard, structured report, or AI-generated output that communicates insights clearly to stakeholders.
- To integrate AI tools (Microsoft Copilot or AI Chatbot) to automate, enhance, or accelerate the analysis and content generation process.
- To document the complete project process, upload all files to GitHub, and record a demonstration video for submission.
- To collect and understand the structure of medical sales data.
- To analyze sales performance using excel formulas and pivot tables.
- To generate meaningful insights that support decision making in the medical sales domain.
- To visualize the full data.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	**Data Collection & Exploration**	The medical sales dataset was collected in excel format. There are 622 entries with order id, order date, item name, price, discount, product segment, quantity, pack name and ect
2	**Data Cleaning & Preprocessing**	Data cleaning was performed using Microsoft excel. Duplicate records were removed, missing values were handled, incorrect format were corrected.
3	**Analysis & AI Integration**	After cleaning, excel formulas and pivot tables were used to analyze data. Formulas like SUMIF, COUNTIF, SUM, MAXIMUM, MIN to see performance of the shop.
4	**Visualization / Output**	Charts such as pie chart, combo charts. And summary tables were created to visually represent sales trends and top performing medical products.

5	**Documentation & Submission**	 • *https://github.com/km1443151-arch/sales-report-of-medical*

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- AI Chatbot (ChatGPT / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and CHATGPT
- Web browser — Google Chrome

B. Dataset

- The dataset used in this project is not publicly available.
- The data was collected from a local medical store sales record.
- The dataset includes information such as medical name, quantity sold, price and total sales.

- AI tools like CHATGPT and copilot were used for guidance in data analysis and project documentation

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

This section summarizes what the team expects to deliver by the end of the project. Outcomes should be realistic and directly tied to the objectives stated above.

- A clean, validated dataset with all preprocessing steps documented and a before/after comparison.
- An interactive dashboard or structured analytical report that clearly communicates key insights from the data.
- Demonstrated use of an AI tool (Microsoft Copilot or AI Chatbot) to generate insights, summaries, or content outputs, with prompts and results documented.
- A GitHub repository containing all project files, a completed fellowship project report, and a video demonstration of the working output.
- Through this project practical skills in data cleaning, analysis and visualization were enhanced.
- It demonstrates how excel data analysis can support better decision making in medical sales management.

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/
- Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com
- Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets
- *https://github.com/km1443151-arch/sales-report-of-medical*